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r2 = res_temp_lookup[i][0];
tempVal = t1 + (t2-t1)*(resVal
- r1)/(r2-r1);
break;
}
}

return tempVal;
}

/*
Main routine begins here
*/

void main( void )
{
    WDCTL = WDTFW + WDTHTL;
    // Stop WDT
    sysOn = 0x01;
    // system in Active mode

    //*****
    * setup BCM (clock module)
    * RSELx = 8, DCOx = 3, MODx = 0; sets
    DCO frequency to 2 MHz
    * Set MCLK & SMCLK on DCO and ACLK on
    internal VLO (~ 10kHz)
    //*****
    DCOCTL = DCO0 + DCO1;
    BCSCCTL = RSEL2 + RSEL1 + RSEL0;
    BCSCCTL3 = LFX1T1SL;

    //*****
    * Setup P2.6 as digital output and p2.7
    as input
    //*****
    P2DIR = 0x40; // p2.6 as o/p
    and p2.7 as input
    P2OUT = 0x00;
    P2IE |= 0x80; // p2.7 as interrupt
    for switch input
    P2IES |= 0x80; // high->low
    transition will trigger interrupt
    P2SEL = 0x00; // configure P2.6
    and P2.7 as digital I/O

    //***** ADC Section *****/

    //*****
    * setup 16-bit ADC
    * Clock source is SMCLK/16 (~125kHz)
    * ADC in low power mode; high impedance
    input buffer on
    * OSR, oversampling ratio = 1024; uni-
    polar result
    * interrupt enable;
    //*****
    SD16CTL = SD16MDIV_2 + SD16LPS +
    SD16SSEL0 + SD16VMIDON + SD16REFON;
    SD16CCTL0 = SD16BUF_1 + SD16CUNI +
    SD16OSR_1024 + SD16IE;
    SD16AE = SD16AEO + SD16AE5;

    //***** PWM DAC Section *****/

    //*****
    * Setup PWM on timer A on TA1 port
    (port 1.2)
    * timer to run off SMCLK @ 2 Mhz
    * timer is in 'up mode' i.e., counts
    upto CCR0
    //*****
    TACCR0 = 1000; // en-
    sures 2 kHz PWM period
    TACCR1 = 0; // ini-
    tialize duty cycle to 0
    TACTL = TASSEL1 + MC0 + TACLK;
    TACTL1 = OUTMOD2 + OUTMOD1 + OUT-
    MOD0; // setup PWM

    varResT = ((68000/65535)*
    NADC)/(2-(NADC/65535));

    // get temperature value from the table
    tempDegC = getTempDegC(varResT);

    // Update pwm value; DMM should display
    temperature in 2V range

    TACCR1 = (tempDegC)/(varSupply);
}

// program comes here after first switch
interrupt service
PISSEL = 0x00; // primary
digital function on all pl ports
PIDIR = 0xFF; // all pl
ports as outputs
PIOUT = 0x00; // all out-
puts set to zero to conserve power
_BIS_SR(LPM1_bits); // enter
sleep mode
_NOP(); // Re-
quired only for C-spy

// program comes here after second
switch interrupt service
// and re-enters the loop
}

// program enters here in temperature
measurement mode
_BIS_SR(LPM1_bits);
// Enter LPM1; SMCLK & ACLK on while
// CPU and MCLK off
_NOP(); // Required only for C-spy

// following is executed after a WDT
interrupt

//*****
* measuring VCC
//*****
SD16INCTL0 = SD16INCH_5;
SD16CCTL0 |= SD16SC; // start
conversion

// wait for the ADC interrupt
_BIS_SR(LPM1_bits);
// Enter LPM1; SMCLK & ACLK ON while
// CPU and MCLK OFF
_NOP(); // needed for debugger

// program enters here after ADC inter-
rupt service
// calculate battery voltage
varSupply = 0.6*(1+NADC/65535);

//*****
* measuring Thermistor resistance
//*****
SD16INCTL0 = SD16INCH_0;
SD16CCTL0 |= SD16SC; // start
conversion

// wait for the ADC interrupt
_BIS_SR(LPM1_bits);
// Enter LPM1; SMCLK & ACLK ON while
// CPU and MCLK off
_NOP(); // needed for debugger

// program enters here after ADC inter-
rupt service
// calculate resistance

```